

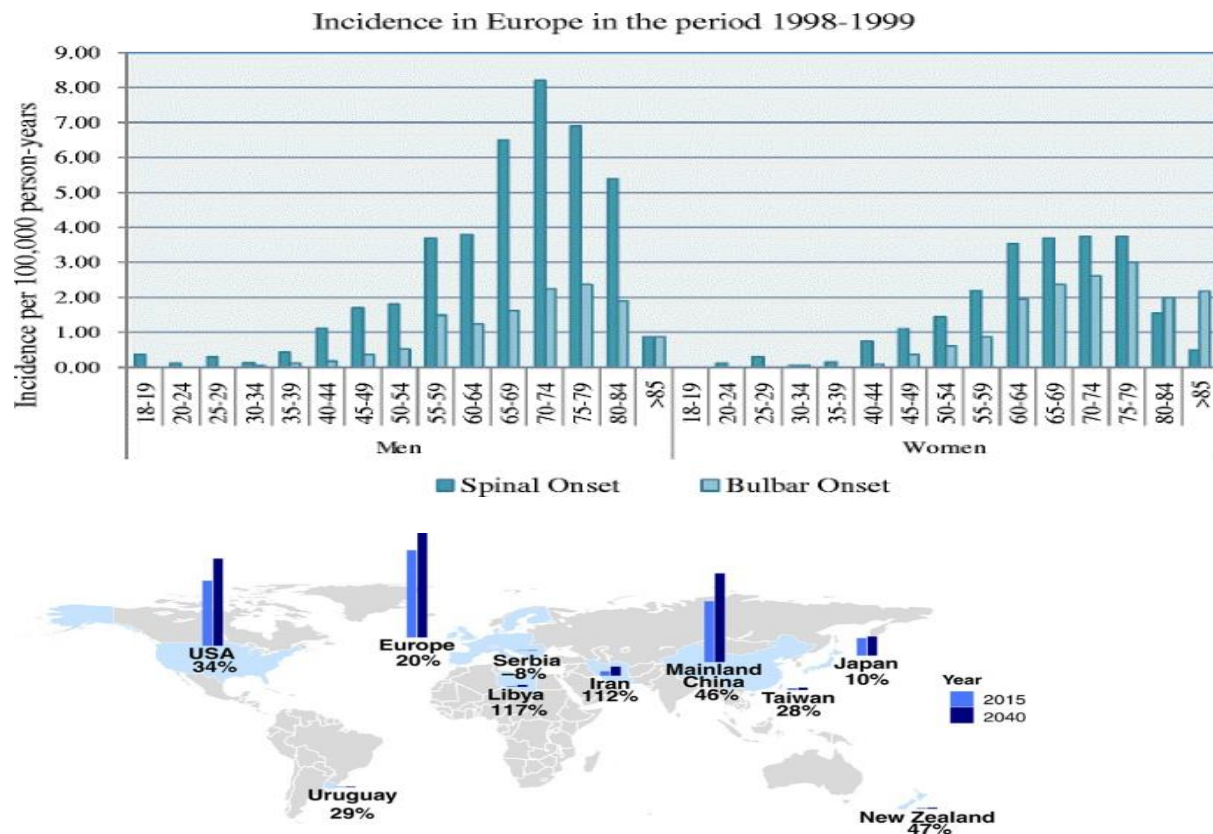
ALS :Amyotrophic lateral sclerosis

1. Definition

ALS is a fatal neurodegenerative disease affecting motor neurons responsible for voluntary muscle movement.

2. Epidemiology

- Incidence: ~2 per 100,000/year
- Usually onset between 55–75 years
- Slightly more common in males
- Average survival: 3–5 years after diagnosis



3. Risk Factors

Non-modifiable

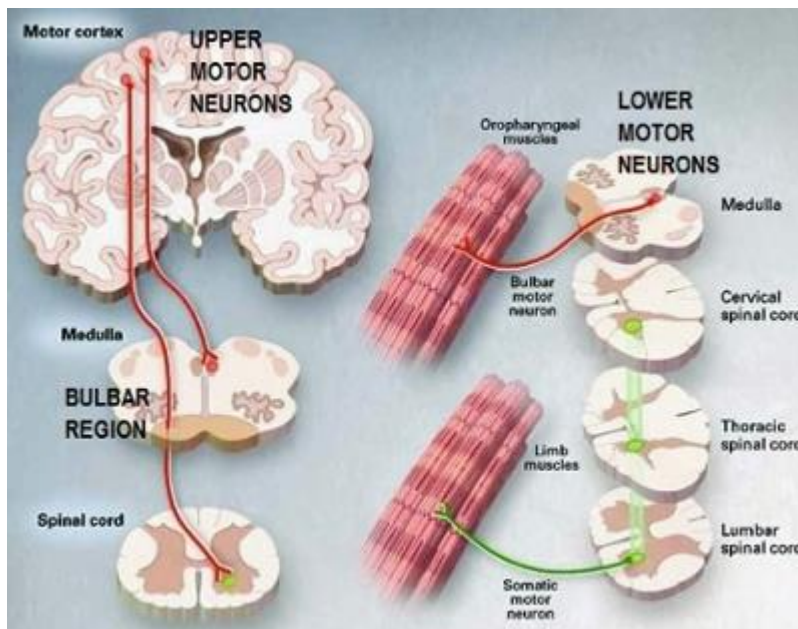
- Increasing age
- Male sex
- Family history
- Genetic mutations

Modifiable

- Smoking
- Pesticide exposure
- Heavy metals
- Military service (association under investigation)

4. Brain Regions Affected

- Primary Motor Cortex
- Corticospinal Tract
- Brainstem Motor Nuclei
- Spinal Cord Anterior Horn Cells
- Corticobulbar Tracts



5. Anatomy

Healthy motor neuron



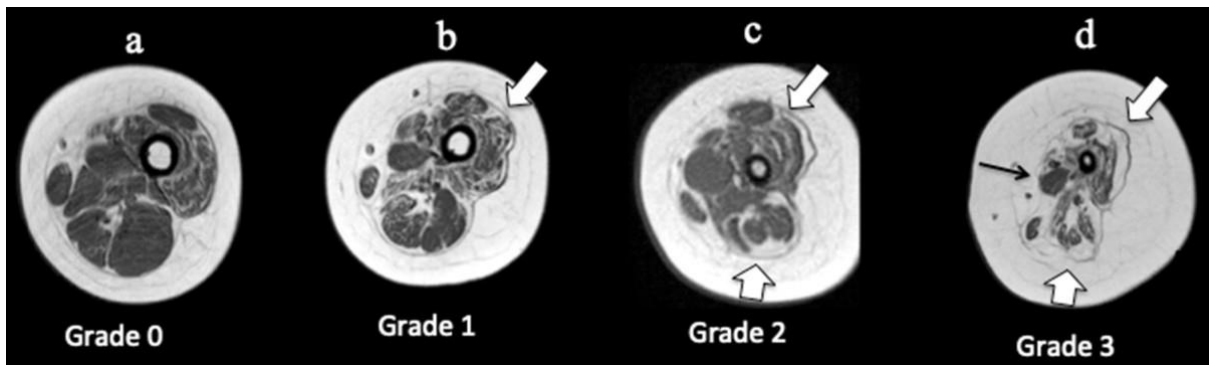
Degenerating motor neuron



Muscle denervation



Muscle atrophy



6. Pathophysiology

Motor Neuron Degeneration

Loss of upper and lower motor neurons

↓

Muscle weakness

↓

Paralysis

Protein Aggregation

- TDP-43
- SOD1
- FUS

↓

Protein misfolding

↓

Neuronal death

Glutamate Excitotoxicity

Excess glutamate

↓

Calcium overload



Motor neuron death

Mitochondrial Dysfunction



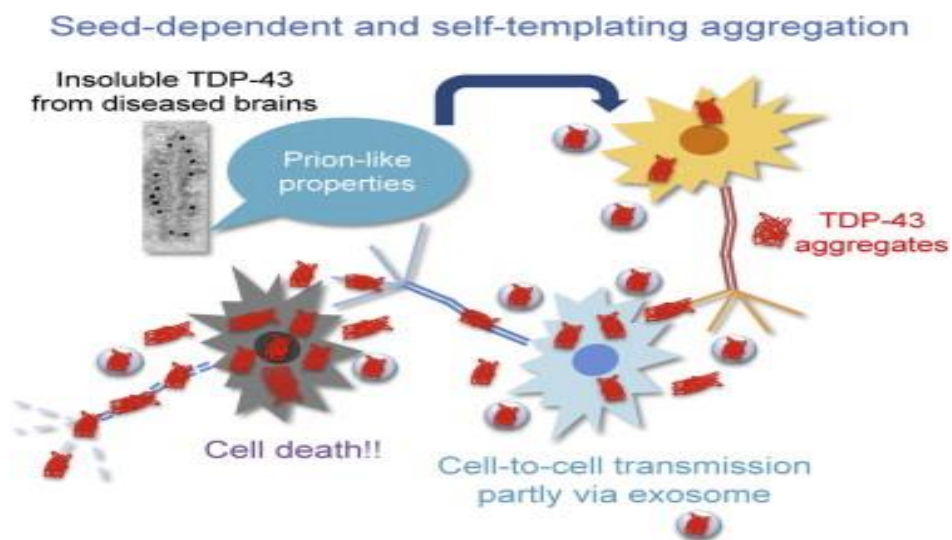
Oxidative stress



Energy failure

Neuroinflammation

- Activated microglia
- Activated astrocytes



7. Genetics

Important genes

- C9orf72
- SOD1

- TARDBP
- FUS
- OPTN

8. Symptoms

Upper Motor Neuron Signs

- Hyperreflexia
- Spasticity
- Babinski sign

Lower Motor Neuron Signs

- Muscle wasting
- Fasciculations
- Weakness
- Hypotonia

Bulbar Symptoms

- Dysarthria
- Dysphagia
- Difficulty chewing

Respiratory Symptoms

- Dyspnea
- Respiratory failure

9. Clinical Stages

- Early ALS
- Regional weakness
- Generalized weakness
- Respiratory involvement
- End-stage disease

10. Diagnosis

- Clinical examination
- EMG

- Nerve conduction studies
- MRI (exclude other diseases)
- El Escorial Criteria

11. Neuroimaging

MRI findings

- Corticospinal tract degeneration
- Motor cortex atrophy

12. Biomarkers

- Neurofilament Light Chain (NfL)
- TDP-43
- CSF biomarkers
- Blood biomarkers

13. Treatment

FDA-approved medications

- Riluzole
- Edaravone

Supportive care

- Physiotherapy
- Occupational therapy
- Speech therapy
- Non-invasive ventilation
- Nutritional support

14. Latest Research

- Gene therapy
- Antisense oligonucleotides
- Stem cells
- AI-assisted diagnosis
- Biomarkers

15. Future Directions

- CRISPR
- Gene replacement
- Neuroprotection
- Personalized medicine
- Digital biomarkers

16. References

- PubMed
- Nature Reviews Neurology
- Neuron
- Lancet Neurology
- Brain
- ALS Association
- WHO